MATH 110 — Week 8 — Solutions

Notes Chapters 14–15; Homework 8

HW8 P1.

Idea. Since $D(x^j) = jx^{j-1}$, divide x^j by $j!$ to remove the coefficient j. The resulting basis has unit chain steps.

1. $D^{n+1} = 0$ and $D^n x^n = n! \neq 0$, so the index is $n+1$. The kernel consists of constants, so the geometric multiplicity is 1.
2. Every nonzero polynomial is a generalized eigenvector for 0, so $G(0, D) = \mathcal{P}_n(\mathbb{C})$ has dimension $n+1$, its algebraic multiplicity.
3. The ordered basis $1, x, x^2/2!, \ldots, x^n/n!$ satisfies $D(x^j/j!) = x^{j-1}/(j-1)!$ for $j \geq 1$ and $D1 = 0$. Hence its matrix is $J_{n+1}(0)$.

HW8 P2.

Idea. The number of blocks gives geometric multiplicity; their total size gives algebraic multiplicity.

Work over $\mathbb{R}$ or $\mathbb{C}$, so $2 \neq -1$. Use

$$A = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix}.$$

1. For (a), $A = \operatorname{diag}(J_3(2), [-1], [-1])$ gives $m_A = (x-2)^3(x+1)$.
2. For (b), $B = \operatorname{diag}(J_2(2), [2], [-1], [-1])$ gives $m_B = (x-2)^2(x+1)$.
3. Each has three coordinates in blocks for 2 and two size-1 blocks for -1; the numbers of 2-blocks are respectively 1 and 2, as required.

HW8 P3.

Idea. Subtracting I leaves the length-3 nilpotent shift.

1. For $N = J - I$, $N(x, y, z) = (y, z, 0)$, $N^2(x, y, z) = (z, 0, 0)$ and $N^3 = 0$. Thus the requested kernels are Fe_1, $\operatorname{span}(e_1, e_2)$ and F^3.
2. $G(1, J) = F^3$, with geometric multiplicity 1 and algebraic multiplicity 3.
3. The binomial formula gives $J^m = I + mN + \binom{m}{2}N^2 = \begin{pmatrix} 1 & m & \binom{m}{2} \\ 0 & 1 & m \\ 0 & 0 & 1 \end{pmatrix}$. Here m and $\binom{m}{2}$ are integer coefficients mapped into F; the formula requires no division by the scalar 2, so also holds in characteristic 2. For $m = 1$ the quadratic coefficient is zero.

HW8 P4.

Idea. Each size-r block contributes $\min(k, r)$ to the kth nullity.

1. For sizes 3 and 2 the nullities at $k = 0, 1, 2, 3, 4$ are $0, 2, 4, 5, 5$.
2. In (b), the space has dimension 5, as specified by the original two block sizes. Nullity 5 at exponent 3 is therefore stabilized. The differences $2, 2, 1, 0$ count blocks of size at least $1, 2, 3, 4$.
3. There are two blocks, both at least 2, and exactly one at least 3, with none larger. Hence the sizes are 3 and 2.

HW8 P5 (optional).

Idea. A repeated nullity fixes the dimension of that generalized eigenspace. Differences count the chains still increasing; their decreases tell us which chains end.

1. For 1, the nullities $0, 1, 2, 2$ have differences $1, 1, 0$, giving one block of size 2. Its generalized eigenspace has dimension 2.
2. For 2, the nullities $0, 2, 3, 4, 4$ have differences $2, 1, 1, 0$, giving two blocks, of sizes 3 and 1.
3. Thus Jordan form is $\operatorname{diag}(J_2(1), J_3(2), J_1(2))$, whose dimensions total $2 + 3 + 1 = 6$.

HW8 P6 (optional).

Idea. The characteristic polynomial counts all generalized directions. To make a power of the shift zero on every block, its exponent must reach the longest chain.

1. $\chi_T = (x-4)^5(x+1)$ and $m_T = (x-4)^3(x+1)$.
2. For 4 the geometric multiplicity is 2 and algebraic multiplicity 5; for -1 both are 1.
3. Blocks larger than 1 occur, so T is not diagonalizable.

HW8 P7 (optional).

Idea. Compute the change in nullity on each block and add.

1. A size-r block contributes $\min(k, r) - \min(k-1, r)$, equal to 1 if $r \geq k$ and 0 otherwise.
2. The kernel of a block-diagonal map is the direct sum of the block kernels, so dimensions add. Summing the contributions gives exactly the number of blocks of size at least k.

Exam practice, Problem 5.

Idea. Triangular diagonal entries give eigenvalues and algebraic multiplicities; the off-diagonal entry at the repeated value determines the minimal exponent.

1. Eigenvalues are 3 and 1, with algebraic multiplicities 2 and 1; hence $\chi_A = (x-3)^2(x-1)$.
2. For $\lambda = 3$, the equations are $y + z = 0$, $-z = 0$, $-2z = 0$, so $y = z = 0$ and $E(3, A) = \mathbb{R}e_1$. For $\lambda = 1$, $2y - z = 0$, $2x + y + z = 0$ give $y = 2t$, $z = 4t$, $x = -3t$, so $E(1, A) = \mathbb{R}(-3, 2, 4)$. The same spanning vectors work over $\mathbb{C}$.
3. $(A - 3I)(A - I) = \left(\begin{smallmatrix} 0 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right) \neq 0$, while $(A - 3I)^2(A - I) = 0$. Thus $m_A = (x-3)^2(x-1)$ by the divisibility criterion and the required root set.

Midterm 2 Review, Problem 8.

Idea. A first kernel counts chains; the minimal exponent measures their largest length.

1. (a) False: $J_3(\lambda)$ has eigenspace Fe_1 of dimension 1, but minimal polynomial $(x - \lambda)^3$, with exponent 3.
2. (b) True in the stated "can depend" sense: λI_2 and $J_2(\lambda)$ have identical diagonals, but minimal polynomials $x - \lambda$ and $(x - \lambda)^2$. Their only differing entry is above the diagonal. This does not assert that every change of an above-diagonal entry changes m_T.